launch-assist-sim

RESEARCH SIMULATOR · GROUND-POWERED LAUNCH ASSIST

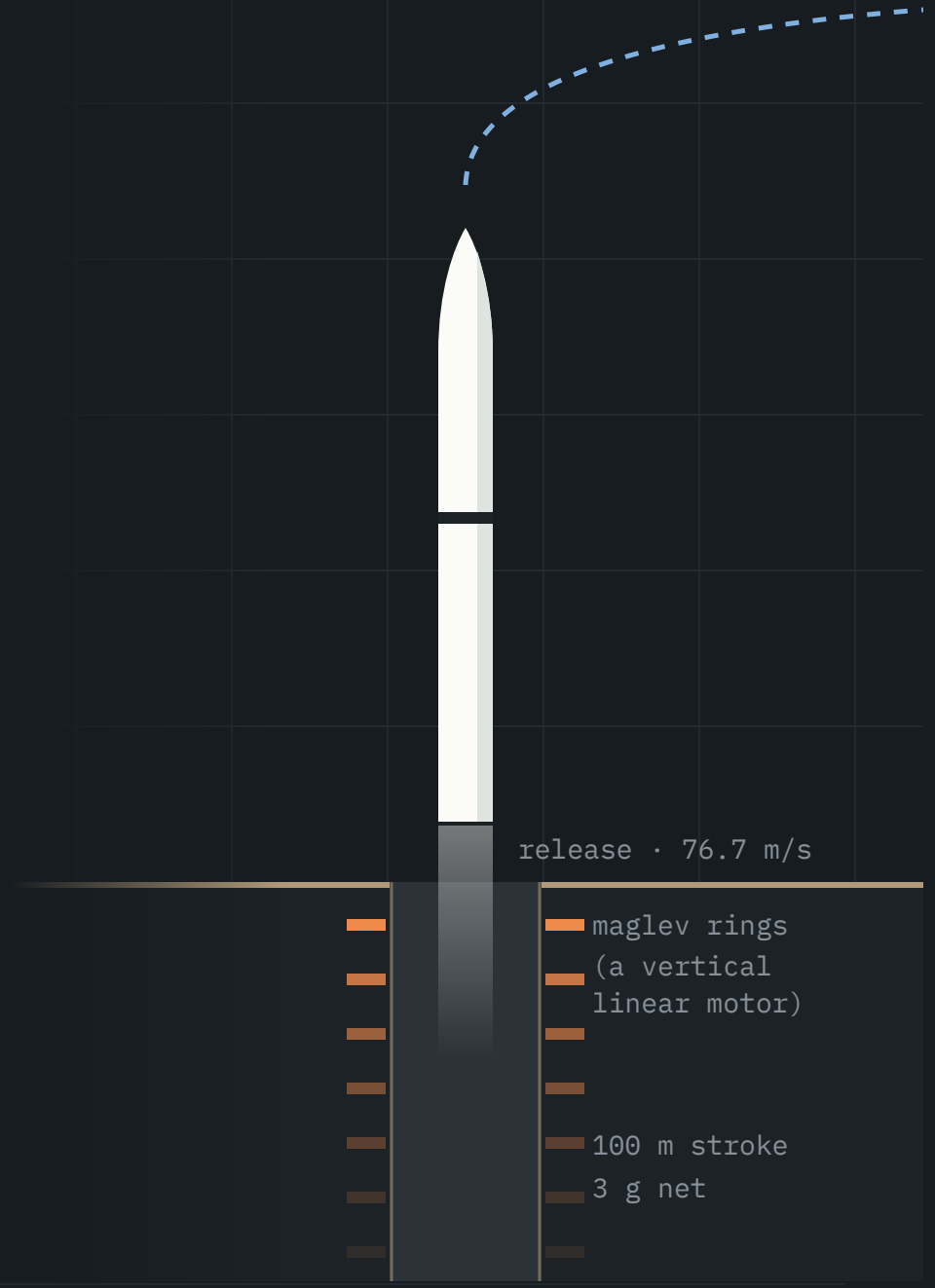
How much rocket propellant can a ground-powered push replace?

A vertical maglev silo pushes a Falcon 9-class rocket out of the ground at about 77 m/s, before or while its engines light. The simulator measures what that is worth, always against the same rocket launched from a pad.

PRELIMINARY RESULTS | PHASES 0-2, SP1, SP2 AND SP7 DONE · SP3 NEXT

STATUS AS OF 9 NOVEMBER 2026

Rahul Singh Khokhar · November 2026



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THE QUESTION

Keep the payload and the orbit. How much propellant can the ground replace?

> **At a fixed payload and the same 200 km circular orbit, what share of a Falcon 9-class rocket's propellant can an electrically powered silo push replace?**

WAY 1 · MORE PAYLOAD

Same propellant, more payload

Measured in Phase 2 on the 2-D model: **+1,498.8 kg** for the cold-start silo (preliminary, slide 10).

WAY 2 · LESS PROPELLANT

Same payload, less propellant

The headline question.
Measured in SP1: **41.26 t** of stage-1 propellant, 10.04% (preliminary, slides 13–15).

CAVEATS Both charge no structure for the 4 g push: ~8.1 t cancels the gain, and SP7's leaves 11.42 t (slide 16). Gate vehicle +14.3% high; as pre-registered, the offload is mostly the lighter stack (slide 14).

WHY A LAUNCH COMPANY MIGHT CARE

- A payload gain helps only on missions that need more mass to orbit. Propellant replaced at a fixed payload would apply at every launch.
- The ground energy can come from renewables and be reused every launch (the hypothesis, README).
- "Energy is cheap; power isn't": the push needs about 1.2 MWh, delivered in 2.6 s at a GW-class peak (README hand calculation).

This is motivation, not a result. Whether any operator wants the trade has not been asked or tested here.

Source: the user's question (docs/handoff/archive, section 1) · README.md · docs/findings/RQ1-fuel-offload-2d.md

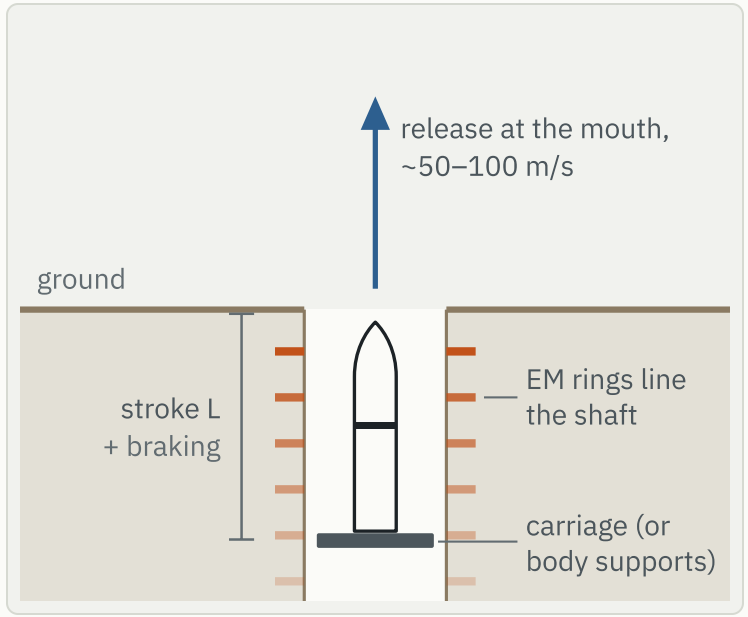
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THE CONCEPTS

Three ways to start: a silo, a ramp, and the pad baseline

MODELLED · CONCEPT A ONLY SO FAR

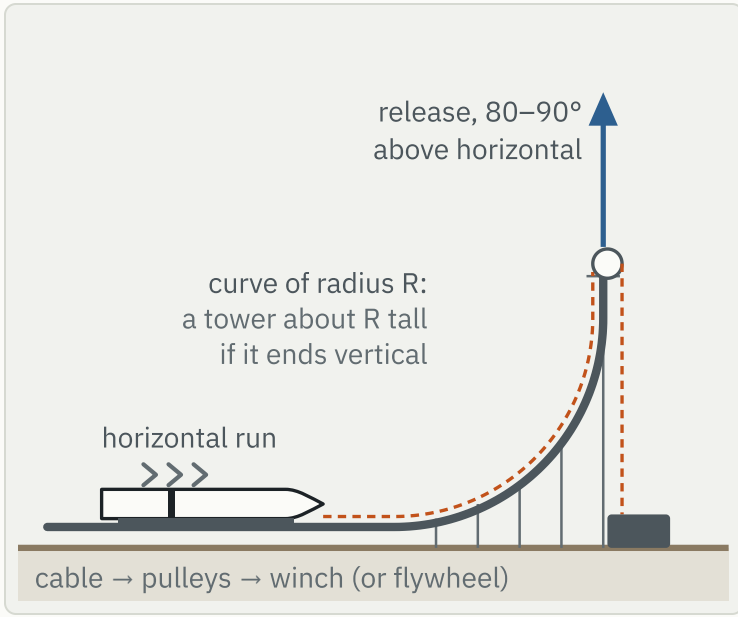
A · VERTICAL MAGLEV SILO



Rings in an underground shaft form a vertical linear motor. Engines light in the shaft (hot) or just after the exit (cold). Modelled as a prescribed-acceleration push and, since SP7, a force- and power-limited linear motor.

PLANNED · README PHASE 3

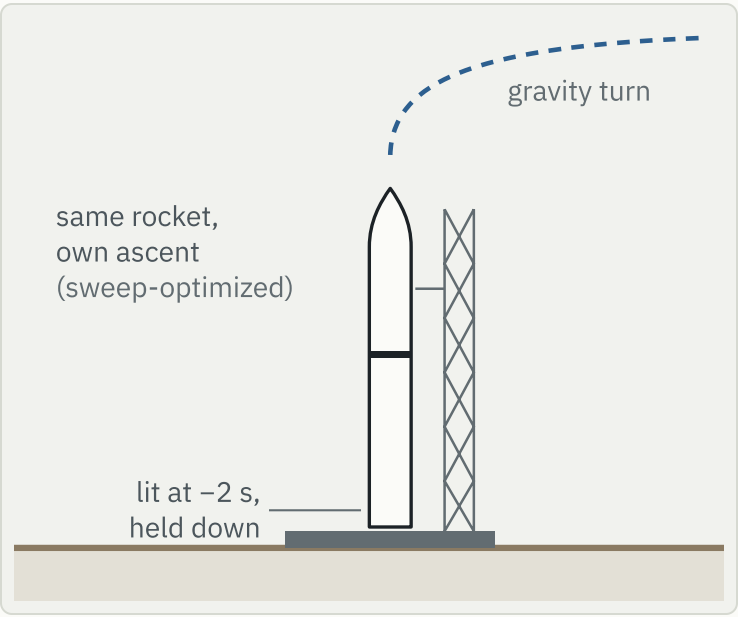
B · CABLE-PULLED RAMP



A winch pulls a carriage along a track that curves from horizontal toward vertical, and the rocket is released at the top. The original idea used a train to pull the cable.

BASELINE · EVERY RESULT IS A DELTA

C · PAD BASELINE



The same rocket from a normal pad, with its own sweep-optimized ascent in the same guidance family. Every result is a difference against C.

Source: README.md, "The concepts" (diagrams redrawn)

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THE HYPOTHESIS

Every metre per second from the ground should help

> **Any ground-supplied velocity, however small, cuts the propellant the rocket needs, or raises its payload. The ground energy can come from renewables and be reused every launch.**
>
> README.md, "The concepts"

HOW IT IS TESTED

- Every run is a difference against the same rocket launched from a pad.
- Pad and variants share the vehicle, the guidance family, the sweep grid and the search budget.
- No physics feature is used in an experiment until its analytic test passes.
- Results that undercut the hypothesis are reported as plainly as the ones that support it.

WHAT COULD SINK IT

- Extra structure for a full, fuelled stack pushed at 4 g felt.
- Speed lost while the engines light after release.
- Aerodynamic loads from starting fast, low in the atmosphere.
- Ground power: gigawatt-class peaks for a few seconds.

"The simulator's real job is to measure these penalties against the ideal numbers" (README).

Source: README.md, "The concepts" and "What can eat the gain" · CLAUDE.md, "Experiments and reporting"

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FIRST-ORDER NUMBERS

About 9 kg of payload per m/s, in the ideal case

HAND CALCULATIONS · MADE BEFORE THE SIMULATOR WAS BUILT

IDEAL ROCKET EQUATION · LOSSES HELD CONSTANT

EXIT SPEED v = √(2aL)

TRACK LENGTH	1 G	3 G	5 G
10 m	14 m/s	24 m/s	31 m/s
50 m	31 m/s	54 m/s	70 m/s
100 m	44 m/s	**77 m/s**	99 m/s
300 m	77 m/s	133 m/s	172 m/s
1,000 m	140 m/s	243 m/s	313 m/s

a is the net acceleration along the track; a vertical silo must also supply 1 g to hold the weight. **Simulation update:** the silo model reproduces √(2aL) exactly, 76.71 m/s after 2.607 s at 3 g net over 100 m (RQ3-1d).

IDEAL SCREENING: WHAT AN ASSIST IS WORTH

ASSIST	PAYLOAD GAIN	OR STAGE-1 PROPELLANT SAVED
25 m/s	+220 kg (+1.0%)	—
50 m/s	+440 kg (+1.9%)	—
77 m/s	+690 kg (+3.0%)	14 t (3.6%)
99 m/s	+890 kg (+3.9%)	18 t (4.6%)
150 m/s	+1,360 kg (+6.0%)	—
300 m/s	+2,790 kg (+12.2%)	53 t (13.5%)

Generic Falcon 9-class two-stage vehicle, 542.6 t (published 549 t); stage-1 dry mass 25.6 t and average Isp 295 s assumed; fairing dropped at staging. NASA's "over 20%" fuel-saving claim for ~275 m/s is a benchmark to explain, not a target. **Simulation update:** at 76.7 m/s the 2-D gate vehicle (+14.3% high) leaves out 41.26 t, 2.76× its own 14.98 t screening offload, before structural mass; part of the beat is the lighter stack and the pad's hold-down burn (slides 13–14).

Source: README.md, "What first-order numbers say" · docs/findings/RQ3-silo-screening-1d.md · RQ1-fuel-offload-2d.md

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WHAT CAN EAT THE GAIN

Nine penalties the simulator has to measure

HAND the README's first-order size · SIM measured since (preliminary, concept A, sweep-optimized, gate vehicle +14.3% high) · also in the README: train traction, sideways loads

AXIAL LOAD ON A FULL STACK

HAND A 3 g net push means a fuelled stack feels 4 g: 21.5 MN at the interface, 2.8× liftoff thrust.

SIM 3.996 g and 22.49 MN on the 2-D stack; SP7 sizes a structure for it (slide 16).

LIGHTING AFTER RELEASE

HAND g × (delay + ramp/2): 5–25 m/s, up to a third of a 77 m/s assist.

SIM 2-D costs about 2× more: 0.5 s + 2 s ramp loses 32.30 m/s on the README masses.

HARDWARE AND STRENGTHENING

HAND Break-even at ~5.5 t added to stage 1, ~0.7 t to stage 2.

SIM ~184 kg of payload per tonne of stage-1 mass, so break-even ~8.1 t; the 41.26 t offload loses 4.5–5.1 t per tonne, zero near 8.5 t. SP7's first-order 6.31 t leaves 11.42 t (slide 16).

PEAK POWER

HAND ≈2.2 GJ in 2.6 s, ~1.65 GW peak; ~1.2 MWh of electricity at 50%.

SIM 2.249 GJ with a 1.725 GW peak on the heavier 2-D stack.

AERODYNAMICS

HAND Extra speed low in dense air raises drag and possibly max-Q.

SIM Unthrottled max-Q −16.0% at full load, +3.4% at the offload; drag loss −3.5 m/s; q-alpha at the kick 1.7–3.4× the pad's.

FAILED IGNITION (A)

HAND An unlit rocket coasts ~300 m up, then falls back into the silo.

SIM Back at the mouth at 76.60 m/s after 15.689 s. No abort modelled.

SILO AIR (A)

HAND A snug shaft is a piston: up to 1.1 MN for a 3.7 m bore, ~5% of the drive.

SIM Vented, no shaft drag (power ~0.08% low); SP7's one sealed shaft adds a column force, still no drag.

CARRIAGE BRAKING

HAND At 77 m/s, braking at 5 g needs ~60 m.

SIM 60.0 m added: the 100 m stroke needs a 160 m facility.

CURVES AND SHALLOW RELEASE (B)

HAND $R = v^2/a_n$: 600 m for 1 g at 77 m/s; an unwinged release droops 5.2°/s at 45°.

SIM Not modelled yet: concept B waits for README Phase 3.

Source: README.md, "What can eat the gain" and its simulation updates · docs/findings (RQ1, RQ2, RQ3, RQ6)

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PRIOR ART

Launch assist is not a new idea

PROJECT	WHAT IT DID	STATUS	LESSON
NASA Magnetic Launch Assist / MagLifter (1990s–2000s)	Horizontal maglev track about 1.5 miles long, 600 mph in 9.5 s	Technology demos only	The closest precedent; its claims are a benchmark to explain
EMALS, U.S. Navy	91 m linear motor launches a 45 t aircraft to 240 km/h; four 121 MJ flywheels	Operational	Proves ~100 m, ~2.5 g, ~70 m/s at 45 t
Holloman maglev sled, USAF (2016)	Rocket-propelled superconducting sled, 633 mph on 2,100 ft	Test facility	Maglev works for sleds near Mach 1
DARPA/NASA Horizontal Launch Study (2011)	Reviewed 130+ horizontal-launch studies spanning 60 years	Only Pegasus became operational	Many concepts close on paper; cost decides
NASA KSC railgun launch assist (2011)	Argued for 2–3 g for several seconds, beyond Mach 1	Analysis, lab tests	Another drive option
Radian Aerospace	~2-mile rocket sled releases a winged spaceplane at Mach 0.7, engines lit	In development; subscale tests 2024	Wings make a shallow release workable
SpinLaunch	33 m vacuum centrifuge, ~10,000 g, 10 test flights (2021–22)	No orbital system built (2025 reports)	Kinetic launch suits only g-hardened payloads
Long March 11, China	58 t solid launcher cold-launched from a tube, igniting after ejection	Operational	"Eject, then ignite" works at 58 t with solids
Intamin Accelerator Coasters (Kingda Ka)	Hydraulic winch and cable: 0–206 km/h in 3.5 s, then a 139 m vertical tower	Kingda Ka demolished 2025	Concept B at park scale, rollbacks included

This project's angle is narrow: measure, on one simulator, what a push is worth to an unchanged two-stage rocket, always against the same rocket from a pad. Pad-start tools (Kerbal Space Program, OpenRocket, RocketPy) cannot ask that question.

Source: README.md, "Prior art" and "Why this exists" (references listed there)

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THE SIMULATOR

Phases joined by events, and every piece tested first

Hold
restrained; engines may start

Assist
1-DOF along the track

Release
mapped into the ascent frame

Ignition
step, ramp or lag

Ascent
1-D vertical or 2-D planar

Staging
dry mass, fairing

Insertion
200 km circular

BUILT IN PHASES 0–2

- 2-D planar ascent over a spherical rotating Earth; a 1-D vertical model.
- ICAO standard atmosphere, drag varying with Mach, back-pressure.
- Vertical rise, kick, gravity turn; linear-tangent steering on stage 2.
- Payload capacity found by bisection; guidance sweep-optimized.
- Constant-acceleration silo push, with no force or power limit (SP7 adds a linear motor, a sealed shaft and a first-order structural sizing).

VALIDATION FIRST

- Rocket equation in vacuum: relative error < 1e-6.
- Elliptical orbit ($e \approx 0.3$), 10 revolutions: energy and momentum drift < 1e-8.
- Equatorial east pad starts at 465.1 m/s inertial.
- Loss budget closes to < 0.01 m/s over a full ascent.
- Assist energy identity < 1e-6; 10× tighter tolerances move payload < 0.1%.

TESTS AND OUTPUTS

200
tests at the Phase 0 gate

269
at the Phase 1 gate

951
928 fast + 23 slow at the Phase 2 handoff

Every run writes `summary.md`, `metrics.json`, CSV time series and plots. `animate` makes an MP4 or GIF; `replay` an interactive HTML page; `scene` the 2-D launch scene as a page; `app` serves the local app with the scene (slide 22).

Source: README.md · CLAUDE.md, "Validation first" · TODO.md (M1, M2) · docs/handoff/archive, section 7

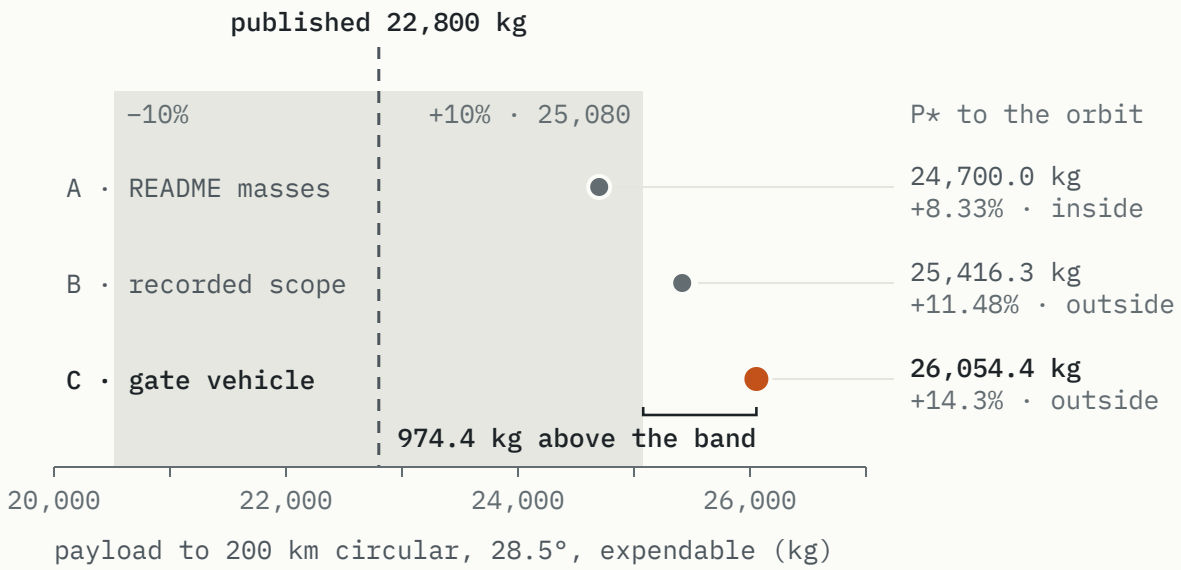
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CALIBRATION

The gate vehicle lands +14.3% high: a documented miss

LABELLED CALIBRATION, NOT VALIDATION

200 KM CIRCULAR · 28.5° · DUE EAST · EXPENDABLE



WHAT IT MEANS The miss carries into every 2-D gain. On a vehicle calibrated to 22.8 t, the cold-start silo's gain would be expected near 1.3 t instead of 1.5 t (derived by proportion, not run).

The reference orbit was fixed before the run and never moved. Nothing was tuned toward the published 22,800 kg.

The pre-registered gate vehicle (mass set C) reaches **26,054.4 kg**, 974.4 kg above the ±10% band's upper edge of 25,080 kg.

Every numerical check passes. The cause is not established, because the omissions point both ways: throttling, reserves and residuals would lower the payload; better guidance could raise it.

The user accepted the miss as documented on 2026-09-30. **Every 2-D number on the next slides is on the gate vehicle and inherits it.**

Source: docs/findings/CAL-f9-leo-2d.md · README.md, "Calibration"

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2-D RESULTS · PHASE 2

To orbit, the cold-start silo gains +1,498.8 kg (+5.75%)

PRELIMINARY FINDING · RQ3-2D, RQ2-2D

CONCEPT A · 3 G NET OVER 100 M · RELEASED AT 76.7 M/S

RUN	STAGE-1 START	PAYLOAD P*	GAIN OVER PAD	
pad (baseline)	lit at −2 s, 2 s ramp, held down	26,054.4 kg	—	
pad_instant (unphysical yardstick)	full thrust at release	26,137.5 kg	+83.1 kg	
silo_instant (unphysical yardstick)	full thrust at release	27,880.4 kg	+1,826.0 kg	
silo_cold (reference)	lit 0.5 s after release, 2 s ramp	27,553.2 kg	**+1,498.8 kg**	
silo_hot_ramp_on_track	lit on the carriage, full thrust at release	27,788.2 kg	+1,733.8 kg	
silo_hot_full	full thrust for the whole push	27,543.0 kg	+1,488.6 kg	

Hatched bars: unphysical yardsticks (a step to full thrust). Bars scale to the largest gain, 1,826.0 kg.

+1,498.8 kg

+5.75% over the pad's 26,054.4 kg

Under every ±10% sensitivity case the gain stays between **+1,345** and **+1,667 kg**.

Best physical variant: lit on the carriage, ramp ending at release, **+235.0 kg** over the cold start, paid for with 2.7 t burned on the track (no sensitivity run).

CAVEATS Sweep-optimized guidance, not optimal control · unthrottled, so max-Q and q-alpha are upper bounds · free kick (no angle-of-attack aerodynamics) · gate vehicle +14.3% high · no structural mass for the 4 g push in these runs (SP7's, slide 16) · prescribed-acceleration drive, massless carriage, no shaft drag · planar point mass, generic C_D(Mach).

Source: README.md, "2-D to orbit" · docs/findings/RQ3-silo-screening-2d.md · RQ2-ignition-timing-2d.md

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2-D RESULTS · THE SCREENING CHECK

Twice the ideal estimate, and the loss breakdown says why

GAIN AGAINST THE IDEAL SCREENING

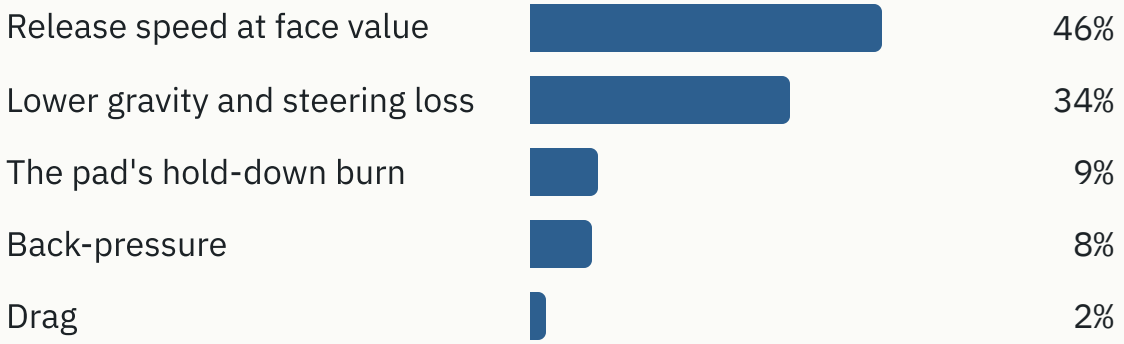


1.96×, about 2.0×. The project's rule: an assist that beats the screening must be explained by the loss breakdown, or it is treated as a bug.

A vertical release is not constant-loss. In silo_cold the gravity, steering, back-pressure, drag and fairing terms together fall by **74.1 m/s**, and the pad pays **14.4 m/s** for burning 2.7 t while clamped. An early vertical m/s is worth about 22.7 kg of payload on this vehicle (release anchor), 2.3× the screening's 10 kg.

BUT SMALL PUSHES gain little from the push itself. At the slow corner of the sweep (0.5 g, 50 m, 22 m/s) the trajectory term is negative (−11.1 m/s), and about 127 kg gross (about 83 kg net, 35%) of that point's +240 kg is the pad's hold-down burn, not the push.

WHERE THE GAIN COMES FROM (MATCHED BASIS)



Capacity basis: 51 / 27 / 10 / 9 / 2%. Nothing is left unexplained beyond search noise.

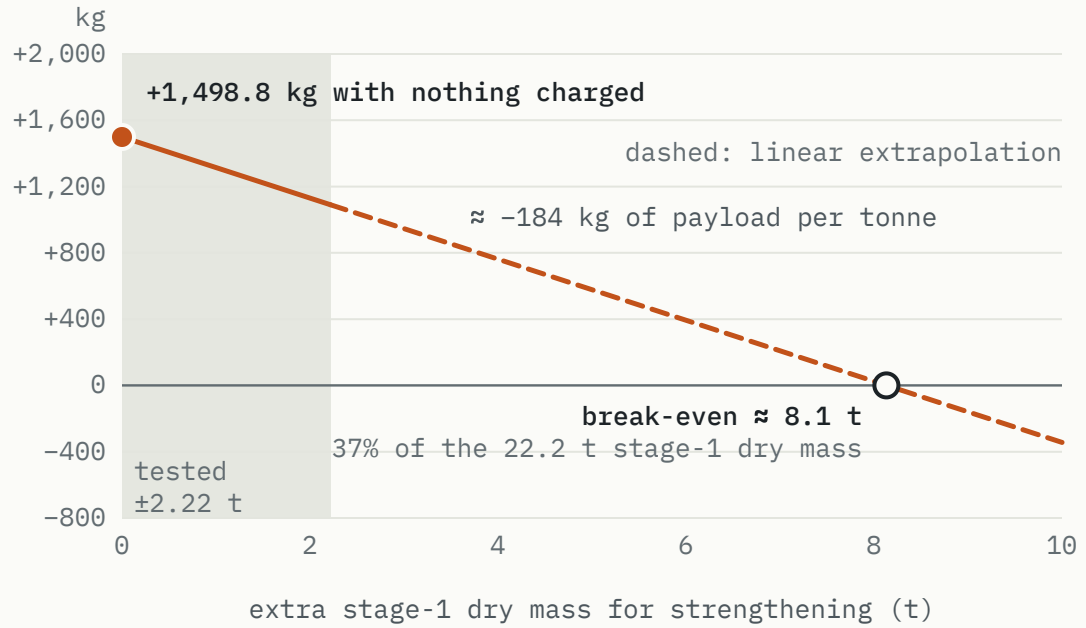
READ WITH The 34% trajectory term appears only after MECO. It is the part a better (Phase 5) pad ascent could change, and optimal control could move it in either direction. The 9% is a baseline convention, not the push.

Source: docs/findings/RQ3-silo-screening-2d.md · README.md, "Why small assists matter" and "2-D to orbit"

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What goes against it, stated as plainly

STRUCTURAL MASS COULD CANCEL THE WHOLE GAIN



Derived from the ±10% stage-1 dry-mass cases of silo_cold; a linear extrapolation beyond them, not a sized structure. These runs charge no mass for the 4.0 g full-stack load (22.5 MN at the interface); SP7's first-order sizing keeps +342 kg of the gain, an upper bound (slide 16).

1. **The startup after release is expensive.** The cold start gives back 327.1 kg (34.74 m/s) against an instant start, 2.37× the 1-D formula. A 3 s lag after the same 0.5 s delay gives back 781 kg.
2. **A full hot start does not pay.** silo_hot_full ends 10.2 kg behind the cold start, and −18.5 to +0.1 kg under the ±10% cases: a tie, not a gain.
3. **Part of the headline is the baseline.** The pad's 2.7 t clamped ramp accounts for 130.7 kg gross (83.1 kg net). Against a pad lit at release the gain is +1,415.8 kg.
4. **The calibration miss inflates it.** On a vehicle calibrated to 22.8 t the gain would be near 1.3 t (derived by proportion, not run).
5. **One aerodynamic load gets worse.** q-alpha at the kick is 1.7–3.4× the pad's (130–251 Pa rad at 77 m/s), with no penalty charged.
6. **There is no abort.** After a failed ignition the full stack falls back to the shaft mouth at 76.6 m/s after 15.7 s.
7. **Small pushes gain little from the push itself.** At 0.5 g over 50 m (22 m/s) the trajectory term is negative (−11.1 m/s), and about 83 kg net (35%) of the +240 kg is the pad's hold-down burn.
8. **The trajectory gain is unverified** beyond the gravity-turn family: it appears only after MECO, and optimal control could move it either way (slide 11).
9. **The facility is sized by power.** A 1.725 GW peak at release after a 2.6 s push, 2.249 GJ per launch, and 160 m of facility with the braking.

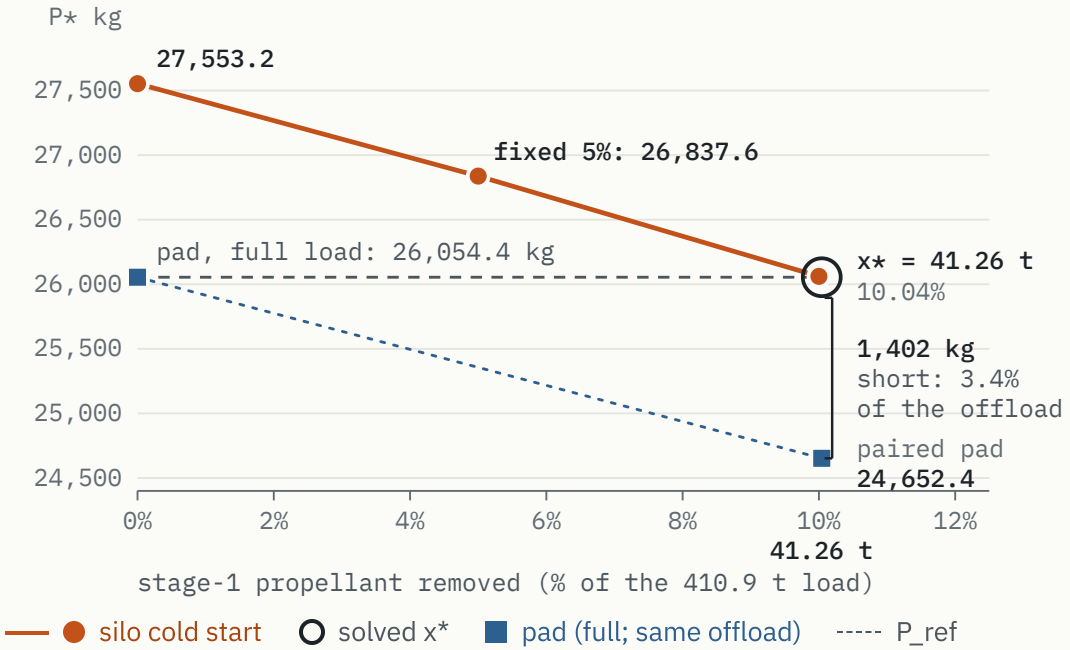
Source: README.md, "Against the hypothesis" · docs/findings: RQ3-2d, RQ2-2d, RQ6

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THE HEADLINE QUESTION · SP1 FINDING

41.26 t less stage-1 propellant, before structural mass

PRELIMINARY FINDING · RQ1-FUEL-OFFLOAD-2D | PRE-REGISTERED · RUN 2026-10-03 | CONCEPT A · 3 G NET OVER 100 M · COLD START



Lines join the recorded runs. The fixed 5% and 10% cases fly their own payload capacity (+783.2 and +6.8 kg over P_ref); x* flies exactly P_ref.

41.26 t

of stage-1 propellant: **10.04%** of the 410.9 t stage-1 load, **7.96%** of the 518.4 t total

- Two different vehicles, the same payload (the pad's 26,054.4 kg) to the same 200 km orbit at 28.5°. Released at 76.7 m/s, lit 0.5 s later, 2 s ramp.
- Checked: the offloaded vehicle's own payload search returns the pad's payload (+0.0085 kg); the pad's own stage-1 control removes nothing.
- ±10% sensitivity arms: **38.67–44.46 t**. 2.76× the ideal-screening estimate (14.98 t); the loss breakdown explains why (next slide).

THE BRIDGE · README MASSES, +8.3%

Calibrated inside the band: **36.01 t, 9.10%** of its 395.7 t stage-1 load. Read it as 9–10% of stage 1 across the two models, not as a Falcon 9 figure.

CAVEATS Gate vehicle, +14.3% high · **as pre-registered, most of the offload is the lighter stack's thrust-to-weight** (the paired pad is only 1.4 t short); a delta-v reading chosen after the run gives the lighter stack 28–29% (slide 14) · sweep-optimized, unthrottled, free kick · no structural mass for the 4 g push (slides 15 and 16) · max-Q 3.4% above the pad's · prescribed 3 g drive, massless carriage, vented shaft · partly filled tanks, dry masses unchanged · stage 1 is the only headline (stage 2 failed its verification).

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THE HEADLINE QUESTION · WHY 2.76× THE SCREENING

Where the 41.26 t comes from, and how much of it is the push

IDEAL DELTA-V THE OFFLOADED VEHICLE DOES WITHOUT

Release speed at face value		+76.7 · 33.6%
Gravity + steering lower losses; only the sum is read		+124.7 · 54.6%
Pad's hold-down burn a baseline convention		+14.4 · 6.3%
Back-pressure		+13.3 · 5.8%
Drag faster in the dense air		−0.9 · −0.4%
Sum (m/s · share)		**228.2 · 100%**

Both vehicles fly the pad's payload; the split closes to 4.5e-12 m/s, so the beat over the screening is explained, not a bug. The offloaded vehicle reaches MECO 12.8 s sooner and loses 1,367.2 m/s to gravity against the pad's 1,492.0.

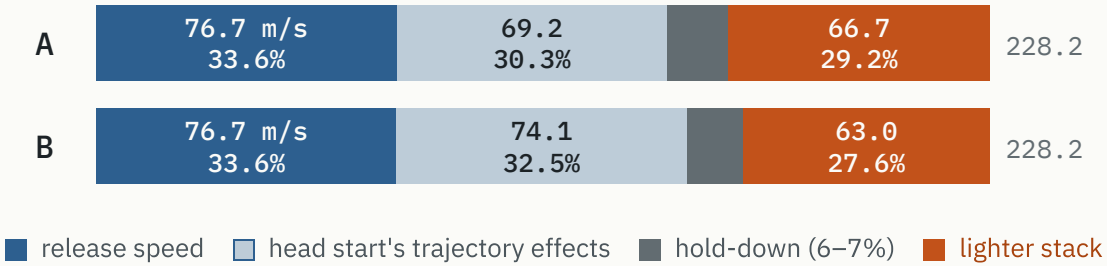
READ WITH The trajectory terms are unverified beyond the gravity-turn family: optimal control (README Phase 5) could move them either way. Fairing −0.02 m/s, margin 0.

HEAD START OR LIGHTER STACK? THE PAIRED PAD

The pad flown with the same 41.26 t offload and no push carries **24,652.4 kg, 1,402.0 kg short**. The push is needed to carry the offload.

AS PRE-REGISTERED That shortfall is 3.4% of the offload, which is small, so the branch fires: **most of the offload is the lighter stack's thrust-to-weight**, not the head start. The note reports this verdict, and argues that it compares payload kilograms with propellant kilograms.

READ AFTER THE RUN In ideal delta-v, a reading the pre-registration does not spell out, the 228.2 m/s splits as follows (two orders of the split):



The lighter stack's own thrust-to-weight gain is 28–29%; the push and its trajectory effects 64–66%. Order A goes through the paired pad, B through the full-load silo. Derived by difference between runs.

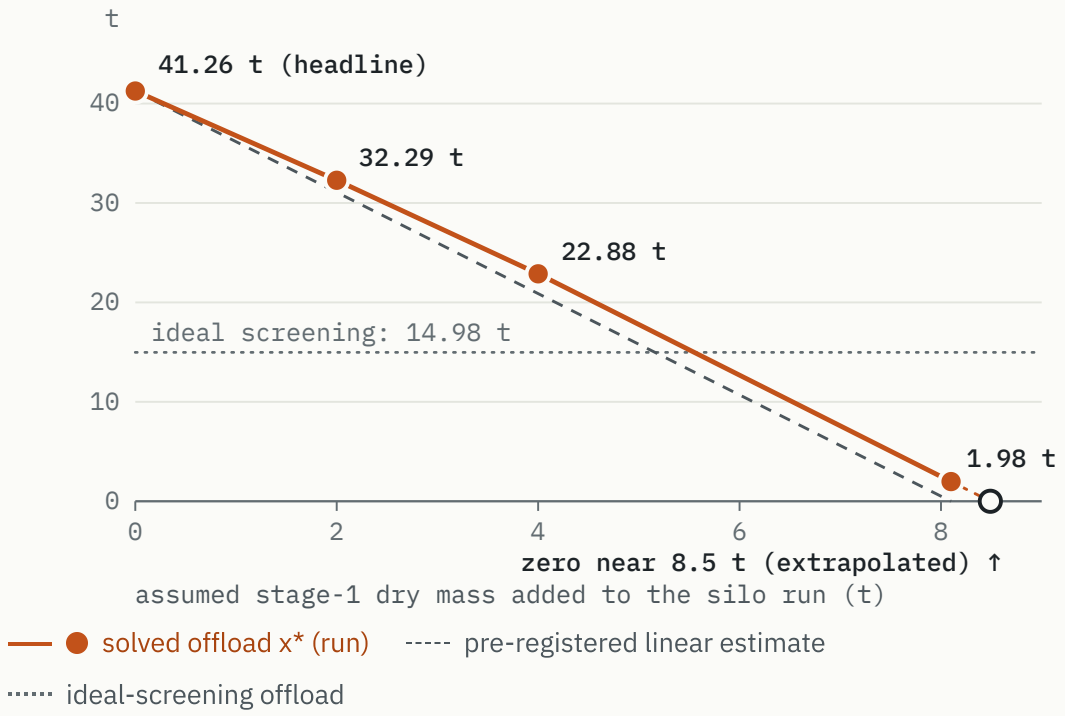
Source: RQ1-fuel-offload-2d.md, "Why about 2.76 times the ideal-screening estimate" and "Head start or lighter stack?"

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THE HEADLINE QUESTION · WHAT CAN CANCEL IT

About 8.5 t of assumed stage-1 structure cancels it

OFFLOAD LEFT AFTER AN ASSUMED STAGE-1 DRY MASS



ASSUMED STAGE-1 DRY MASS	OFFLOAD LEFT	% OF STAGE 1	MAX-Q VS PAD
none (headline)	41.26 t	10.04%	+3.35%
+2 t	32.29 t	7.86%	−2.18%
+4 t	22.88 t	5.57%	−7.57%
+8.1 t	1.98 t	0.48%	−18.19%
≈ 8.5 t (extrapolated)	0	0	—

Each tonne takes 4.5–5.1 t off, and the erosion steepens; every row lies above the pre-registered linear estimate. 8.5 t is 38% of the 22.2 t stage-1 dry mass.

- **Loads at the headline.** Max-Q 38.4 kPa, 3.35% above the pad's: the full-load silo's 16.0% reduction is spent. q-alpha at the kick is 3.0× the pad's, and the stack feels 4.0 g on the track (20.81 MN). Unthrottled.
- **Energy, not an efficiency claim.** The 12.4 t of RP-1 removed carry about 128× the push's 1.16 MWh of electricity in combustion heat (5.65× at +8.1 t); the ratio leaves out LOX production, fuel supply, grid losses and the facility.
- **Stage 2 is not a headline.** Its case failed independent verification (31.90 t gross, a flagged lower bound).

ASSUMPTIONS The rows add an assumed dry mass to the silo run only; they are not a sized structure. SP7's first-order sizing replaces the assumption with a modelled mass: 6.31 t, 11.42 t left (next slide, with its caveats).

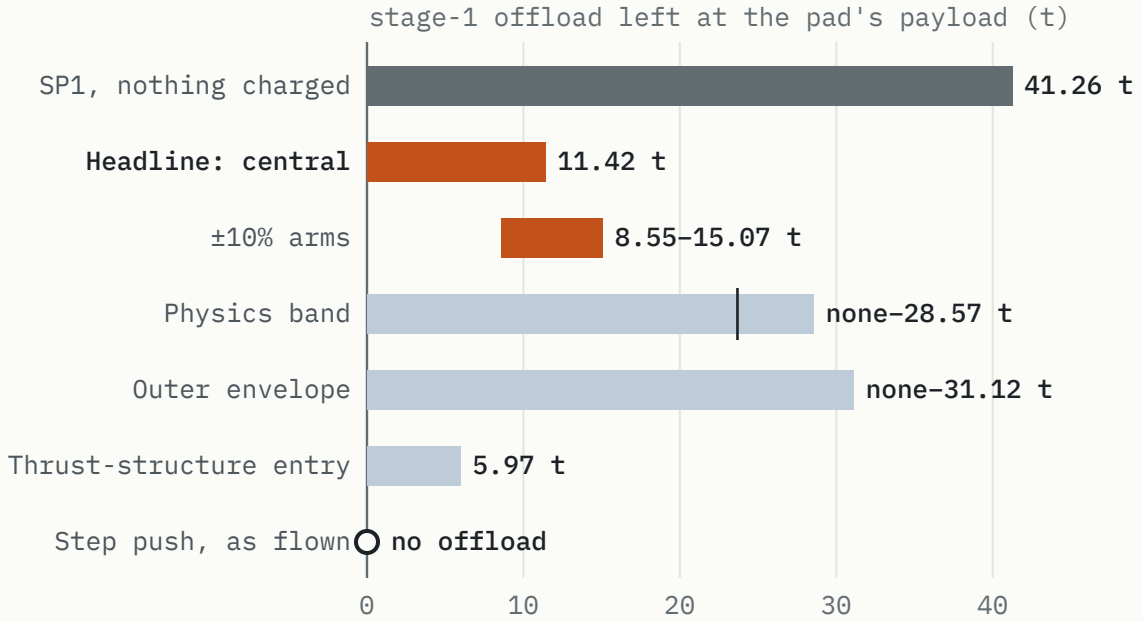
Source: docs/findings/RQ1-fuel-offload-2d.md, "Structural penalty rows and break-even", "Energy" and "Max-Q and loads"

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THE HEADLINE QUESTION · SP7 FINDING

With a sized structure charged, 11.42 t of the 41.26 t is left

PRELIMINARY FINDING · RQ1-STRUCTURAL-2D | PRE-REGISTERED · RUNS FROM 2026-10-16 | FIRST-ORDER SIZING, NOT A STRUCTURAL DESIGN



The tick on the physics band is 23.68 t, before its plausibility scaling. "None": the case falls short of the pad's payload with nothing offloaded. The bands are the extremes a screened search found, not bounds. The 5.97 t sits close to its own estimated break-even, and k_ts's central moved after the planning probe in a direction that raised it (the note).

11.42 t

27.7% of SP1's uncharged 41.26 t, **2.78%** of the 410.9 t stage-1 load: the pre-registered reading "cancels most" fired

- A structure is sized for the offloaded stack and charged to the assisted run at every offload the solve tries. Its **6.31 t** costs **29.84 t** of offload, about 4.7 kg per kg (derived).
- No offload at the physics band's high-mass end (243.7 m/s short, −2,080 kg), the outer envelope's (1,021.5 m/s, −7,804 kg) or the step push the constant-acceleration drive actually flies (125.2 m/s, −1,087 kg). The −2,080 and −1,087 kg are upper bounds on the payload change.
- Sized for its own full-load push, silo_cold keeps **+342 kg** of its uncharged +1,498.8 kg, an upper bound: its kick's q-alpha is above the pad's.
- README-loads fork (+8.33%): **9.16 t** of its 36.01 t, 25.5%.

CAVEATS First-order station sizing at central coefficients · an assumed 0.5 s rise time the constant-acceleration trajectory does not fly · zero existing margin, the pad's own unthrottled flight as the envelope · the instant release flagged, not sized (the model's single-mode step separates the LOX column, as it does stage 2's at the pad's own MECO) · gate vehicle +14.3% high · sweep-optimized, unthrottled, free kick (q-alpha 1.89× the pad's) · partly filled tanks.

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At 3 g, 25 and 50 m strokes leave none; more stroke keeps more, at a price

SAME 3 G PUSH, A LONGER STROKE EXITS FASTER

STROKE	EXIT	UNCHARGED	CHARGED	FACILITY
25 m	38.35 m/s	18.77 t	none (60.2 m/s short)	40 m
50 m	54.24 m/s	28.65 t	none (18.3 m/s short)	80 m
100 m	76.71 m/s	41.26 t	11.42 t	160 m
200 m	108.48 m/s	56.83 t	31.62 t	320 m
300 m	132.86 m/s	67.29 t*	44.73 t*	480 m

* An upper bound: an unconstrained kick above 120 m/s. Peak drive power 2.44 and 2.99 GW at 200 and 300 m (full-load runs).

- **The lowest exit speed that still offloads** lies between 54.24 and 76.71 m/s; the sweep does not resolve the crossing. Below it the extra speed helped, but did not pay for the structure it needs.
- **Above 100 m, faster exits kick harder.** The 200 and 300 m full-load runs kick into 574.1 and 1,293.3 Pa rad, 7.7 and 17.3× the pad's, and the structural runs' values, not written, are likely higher. Nothing constrains it.

SAME 76.71 M/S EXIT, A LONGER STROKE PUSHES GENTLER

STROKE	FELT (FLOWN)	STRUCTURE	CHARGED	FACILITY
50 m	6.996 g	16.05 t	none (190.0 m/s short)	110 m
100 m	3.996 g	6.31 t	11.42 t	160 m
200 m	2.496 g	3.20 t	26.69 t	260 m
300 m	1.996 g	2.06 t	32.03 t	360 m

Uncharged, every stroke offloads the same 41.26 t in this model. The 50 m push sizes stage 2 too.

- **Depth sets the felt load, and the felt load sets the mass:** the gentler push needs less structure, for 1,769.8 kWh at 300 m against 1,235.1 kWh at 100 m.
- **The returns fall.** The step from 100 to 200 m adds 15.27 t; the last 100 m adds 5.34 t.

CAVEATS Central structure only; a sweep point has no flown band and no ±10% arm · every point flags the instant release and the payload limit · a deeper or faster silo is a larger facility, and no facility cost is modelled · first-order sizing, gate vehicle +14.3% high, sweep-optimized, unthrottled, free kick.

Source: RQ1-structural-2d.md, both axis sections · sweep 20261019T172327Z (re-run; git record dirty, deviation 45)

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THE HEADLINE QUESTION · NINE LINEAR-MOTOR DRIVES

All nine drives cancel most of it; on the same silo, neither faster exit keeps more

DRIVE (SAME 100 M SILO)	WHAT IT CHANGES	EXIT AT X* (M/S)	OFFLOAD LEFT
lm_cmd_hot	acceleration law, lit 0.849 s into the push	76.71	18.27 t
lm_fixed_sled22	force law, 22 t carriage	75.27	11.64 t
lm_cmd	acceleration law: a flown ramp	76.71	11.42 t
silo_cold	constant_accel, the basis	76.71	11.42 t
lm_fixed	force law for the full load	77.15	11.34 t
lm_fixed_target	force law, released at 76.71 m/s	76.71	11.05 t
lm_fixed_sealed	force law, sealed shaft	72.22	9.05 t
lm_fixed_hot	force law, lit at push start	89.24	7.35 t
lm_fixed_r07	12% more force, power-limited	76.60	5.66 t
lm_cmd_taper	acceleration law, stop taper	76.71	2.48 t

Each variant with the central structure charged, at its solved offload x*, where it carries the pad's payload. The force law's stack is lighter or heavier than the full load at x*, so its exit speed moves; the sealed column and on-track thrust move it too. lm_cmd equals silo_cold, and lm_cmd_hot the constant-acceleration ramped hot start, by construction, not as evidence about a real drive.

- **Two exit faster than silo_cold at x*, and neither keeps more.** lm_fixed_hot exits 12.53 m/s faster and keeps 4.07 t less; lm_fixed 0.44 m/s faster and 82 kg less. A longer stroke's faster exit does keep more (slide 17).
- **A gentler release costs offload and buys only flags.** The stop taper keeps 8.94 t less; the model sizes no release step, so its benefit shows only in the flags.
- **A slower exit can leave more.** The 22 t carriage exits 1.43 m/s slower and keeps 0.22 t more: it lowers the felt load. The carriage itself is not sized.
- **The air column costs too.** The sealed shaft keeps 2.29 t less than the same drive vented, at a 5.24 m bore, under a uniform adiabat.
- **The best is a hot start.** lm_cmd_hot keeps 18.27 t, 44.3% of the 41.26 t, still under half, favoured by the unmodelled exhaust in the shaft and with its release flag, a risk no number here measures.

CAVEATS Central structure only, with no physics band and no ±10% arm: a difference of 0.22 t or 82 kg is read there and nowhere else · force and power caps only, no thermal, current or voltage limit · an assumed efficiency of 0.5 · a massless carriage except one · an instant release, flagged · the exhaust in the shaft is not modelled · every caveat of slide 16 applies.

Source: RQ1-structural-2d.md, "The drive cases" · run 20261103T231946Z (commit 642e11e)

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AGAINST THE HYPOTHESIS · THE OFFLOAD WITH ITS STRUCTURE (SP7)

Not every m/s of assist pays once its structure is charged

1. **The structure cancels most of the offload** at the central coefficients: 11.42 t of 41.26 t is left (every caveat of slide 16 applies).
2. **It no longer beats the ideal-screening estimate:** 0.763 of the 14.98 t at 76.7 m/s, where SP1's uncharged headline was 2.76×.
3. **The physics band reaches no offload:** 243.7 m/s short and −2,080 kg at its adverse end, a bound on the payload change.
4. **The push the constant-acceleration drive actually flies cancels all of it.** The instant step leaves no offload; a rise time at or below about 0.14 s is estimated (sizing only) to do the same.
5. **Short strokes give none at all:** 25 and 50 m at 3 g, the same felt load and nearly the same structure for a smaller offload.
6. **A fast, short push sizes the upper stack:** at 7 g, 16.05 t of structure and 190.0 m/s short.
7. **Every offloaded structural run kicks above the pad's q-alpha,** 1.7 to 3.9× on the recorded constant-acceleration runs (the sweeps' structural runs are not written), 1.39 to 5.48× on the drive cases, a lateral load the pad envelope does not cover.
8. **Other plausible design choices leave little:** 5.97 t through the thrust structure (close to its own estimated break-even), 4.81 t with the envelope capped at 4.0 g.
9. **The force-limited hot start loses:** 3.99 t less than its own cold start, 4.07 t less than silo_cold (the rise formula overcharges its n_peak, which costs of order 1 t of offload on a rough estimate, not measured; the reading does not move).
10. **Most unmodelled terms favour the assist:** the unthrottled envelope, the release not sized, propellant settling for the ignition 0.5 s after a release the model flags as separating the LOX column (a risk no number here measures), COPVs and engine mounts not charged.

IN THE ASSIST'S FAVOUR, AS PLAINLY

- The low-mass ends keep more than half: 28.57 t (23.68 t unscaled), 31.12 t on the outer envelope.
- A deeper silo at the same exit speed keeps 32.03 t at 300 m, 2.0 g felt, for 360 m of facility.
- The two constant-acceleration hot starts keep 4.9 to 6.9 t more than the cold start, with the exhaust caveat favouring them and with their release flag, a risk no number here measures; lm_cmd_hot repeats the 6.85 t by construction (it flies the plateau that case's structure is sized for, with the same thrust ramp and exit speed).
- Some omissions and choices work against the assist: zero existing margin, a clean-sheet vehicle, damping.

WHAT WAS KNOWN BEFORE THE RUN The pre-registration's sizing-only estimates placed the headline at 11.1 t and the band from no offload to 28.5 t, and the central values and range ends were set after the planning probe had shown which coefficients matter. What was fixed in advance is the design around these numbers and how each result is read.

Source: docs/findings/RQ1-structural-2d.md, "What goes against the hypothesis" and "Caveats (read first)"

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ROADMAP

One phase per session, each ending in something you can run

DONE	DONE	DONE	DONE · 4 OCT	DONE · 7 OCT	DONE · 9 NOV	NEXT	PLANNED	PLANNED	PLANNED
Phase 0	**Phase 1**	**Phase 2**	**SP1**	**SP2**	**SP7**	**SP3**	**SP4**	**SP5**	**SP6**
Scaffold, atmosphere, configs, CLI	Vertical 1-D model and the silo push	2-D ascent to orbit; calibration miss accepted	Launch settings and fuel offload at fixed payload (planar)	Local app with a 2-D launch scene (slide 22)	Structural mass for the 4 g push; force-limited drive (slides 16–19)	3-D dynamics: point mass over a rotating sphere	3-D scene; headline re-checked in 3-D	Oblate Earth: J2 and the ellipsoid	6-DOF verification fly-out

README PHASE 3 · ASSIST MODELS

SP7 built and ran the first-order structural sizing, the force- and power-limited linear motor, carriage mass, braking functions and the sealed shaft: 11.42 t of the 41.26 t is left at the central structure, with its findings note; the phase closed on 9 November 2026. Curved ramp and cable winch are not scheduled; the 3-D phases come next, from SP3.

README PHASE 5 · FAIR COMPARISON

Optimal-control ascent for every configuration; the headline results re-run with optimized guidance instead of sweep-optimized.

README PHASE 6 · HOBBY SCALE

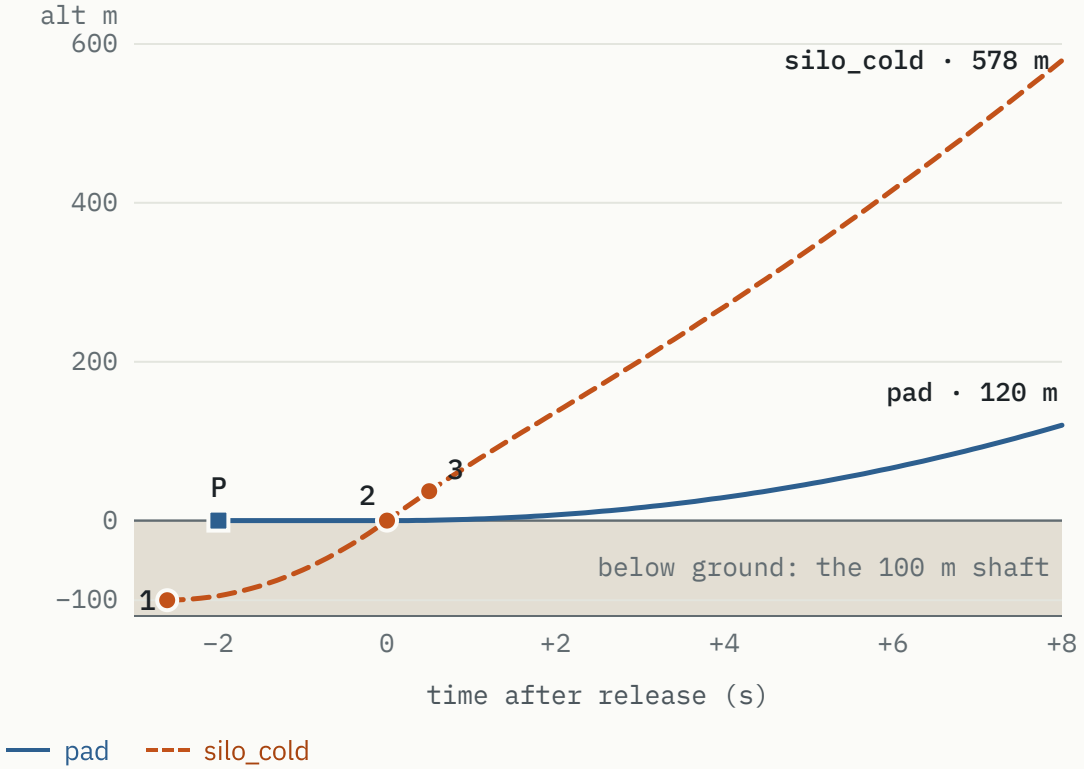
A RocketPy 6-DOF model of a real rocket with an assist before the rail; baseline apogee within ±5% of RocketPy.

Source: docs/phases/README.md (program board) · README.md, "Roadmap" · TODO.md

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SEE IT FLY

The head start, in the first 8 seconds of the recorded run



1. The push starts 100 m down: 3 g net, 4.0 g felt.
2. Release at the silo mouth at 76.7 m/s.
3. Lit 0.5 s later, 37 m up, with a 2 s ramp.

P. The pad: lit at −2 s and held down until release.

At 8 s after release the silo stack is at 578 m and 85 m/s; the pad at 120 m and 31 m/s.

ANIMATION EXAMPLES

- **SP7 structure charged**: pad against silo, replay, video and scene.
- **SP1 fuel offload**: against the pad and its paired pad, and SP2's launch scene from the recorded directory (the app: next slide).
- **Pad against cold-start silo**: both flown to a 200 km orbit.
- **Ignition timing**: four ways to light stage 1.
- **Failed ignition**: the push with no engines, back at the mouth at 15.7 s.

Videos, GIFs, replays and scenes: khoks.github.io/.../examples/

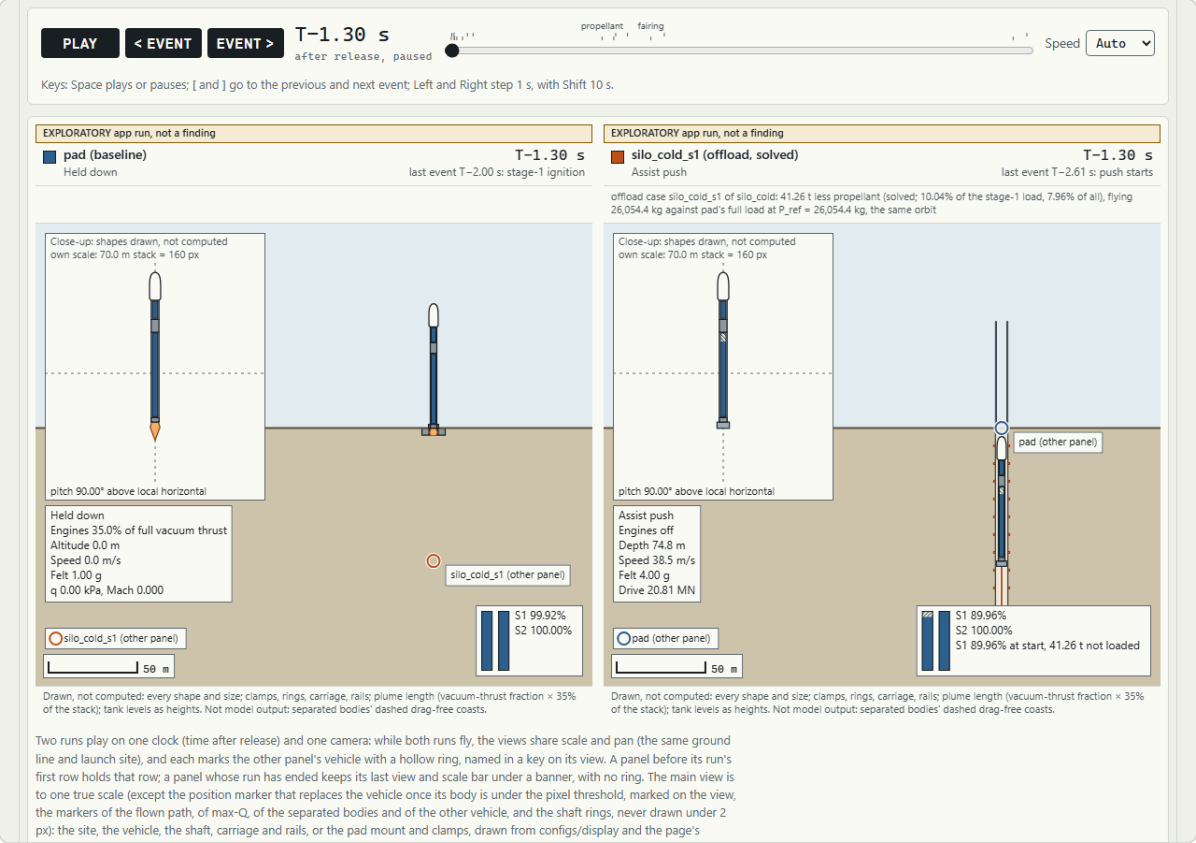
Point-mass planar model, sweep-optimized and unthrottled, gate vehicle +14.3% high, no structural mass for the 4 g push in this run. A model replay, not a design result.

Source: results/silo_screening_2d/20260930T175743Z, pad and silo_cold timeseries.csv and events.csv · site/examples/

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Set the silo, press Launch, watch pad and silo side by side

EVERY APP RUN EXPLORATORY · NEVER A FINDING | LOCAL ONLY · 127.0.0.1 | DONE · 7 OCTOBER 2026



Exploratory, not a finding. An app screenshot from the recorded demo (silo_cold_s1 preset) at T−1.30 s: left the pad held down, right the silo run on its carriage with 41.26 t not loaded (the hatched band on its stage-1 gauge). The banner on each panel marks every app run.

- `launchsim app` : a form built from the committed experiments (depth, push, ramp start, startup, offload, penalty; 17 presets), Launch, a progress card, a results panel with every caveat beside every number, a browser over the results on disk, the scene, an MP4 export.
- `launchsim scene` : the same scene as a standalone page. The vehicle's path, thrust and mass are replayed from the recorded time series; the spent stage, the fairing halves, the carriage after release and the unpowered attitude are display-only reconstructions, named on the page and on every video frame.
- Every launch writes `results/app/` , labelled exploratory with a banner and no plots; findings come only from committed experiment files run from a clean tree (one disclosed departure, which reached no number: SP7's sweep re-run, RQ1-structural-2d.md, Provenance). No physics changed in SP2.

REPRODUCTION, NOT EVIDENCE The app's launch of its `silo_cold_s1` preset gives SP1's 41.26 t and the pad's 26,054.4 kg within 0.002 kg: the same configuration on the same model, so a reproduction of slide 13, not new evidence, with all of its caveats (gate vehicle +14.3% high, no structural mass for the 4 g push, sweep-optimized and unthrottled).

Source: docs/demos/SP2/README.md (launch 3) · docs/manual/07b-app.md · docs/phases/SP2-launch-app-2d-scene.md

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READ MORE

Where the full numbers and their caveats live

ON GITHUB

github.com/khoks/RocketLaunchOptimizationSimulator

README.md: the research brief, prior art, hand numbers and results so far.

docs/findings/: one note per research question, caveats first.

docs/phases/: the program board, SP1 to SP7; **docs/demos/**: the recorded demo of each closed phase.

ON THE SITE

khoks.github.io/RocketLaunchOptimizationSimulator

User manual: install, run, the app and the scene, experiment and vehicle files, outputs.

Animation examples: replays and the launch scene of the recorded 2-D runs.

This deck, also as a PDF.

HOW IT IS RUN

From the user manual (Python 3.12 and uv):

```
uv sync
```

```
uv run python -m launchsim run
experiments/silo_screening_2d.yaml
```

```
uv run python -m launchsim replay
results/silo_screening_2d/<timestamp>
--runs pad silo_cold
```

```
uv run python -m launchsim app
```

Reading is free. Any other use needs written permission (next slide).

FINDINGS: THE SUGGESTED READING ORDER

CAL-f9-leo-2d
The vehicle every 2-D number rests on, and its +14.3% miss.

RQ1-fuel-offload-2d
Propellant replaced at fixed payload, with penalty rows.

RQ1-structural-2d
The same offload with a sized structure charged.

RQ3-silo-screening-2d
The payload gain and the full caveat list.

RQ2-ignition-timing-2d
Hot or cold start; delay, ramp and lag.

RQ6-aero-2d-preliminary
Max-Q, q-alpha and drag, unthrottled.

RQ3-1d, RQ2-1d
The 1-D record: closed forms, loads, energy.

Source: README.md, "Quick start" · docs/manual/ · docs/findings/README.md

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- You may view the code, documentation, findings, images, animations and slides on GitHub.
- You may not copy, modify, merge, publish, distribute, sublicense, sell or otherwise use any part of it, commercially or not, without prior written permission.
- The research ideas, including the vertical maglev silo push and the fixed-payload propellant-replacement analysis, are disclosed for reading and discussion. Publication grants no right to implement or commercialise them. Earlier launch-assist work (NASA MagLifter, EMALS, Long March 11 and others) is credited on the Prior art slide (7).
- Results are preliminary and are not engineering advice. Permission requests: through GitHub, @khoks.

github.com/khoks/RocketLaunchOptimizationSimulator